FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Advanced), 2015

PART TEST – IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 198**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 08)** contains 8 multiple choice questions which have only one correct answer. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (09 to 14) contains 3 paragraphs with each having 2 questions. Each question carries **+3 marks** for correct answer and **– 1 mark** for wrong answer.

Section-A (15 to 20) contains 6 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

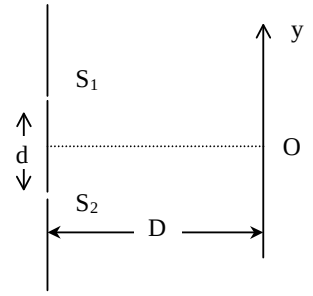
Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Two mirrors, one concave and the other convex, are placed 60 cm apart with their reflecting surfaces facing each other. An object is placed 30 cm from the pole of either of them on their axis. If the focal lengths of both the mirrors are 15cm, the position of the image formed by reflection, first at the convex and then at the concave mirror, is
   (A) 19.09 cm from the pole of the convex mirror
   (B) 19.09 cm from the pole of the concave mirror
   (C) 11.09 cm from the pole of the concave mirror
   (D) 11.09 cm from the pole of the convex mirror

2. An alternating voltage $E = E_0 \sin \omega t$, is applied across a coil of inductor L. The current flowing through the circuit at any instant t is
   (A) $\frac{E_0}{\omega L} \sin(\omega t + \pi/2)$
   (B) $\frac{E_0}{\omega L} \sin(\omega t - \pi/2)$
   (C) $E_0 \omega L \sin(\omega t - \pi/2)$
   (D) $E_0 \omega L \sin(\omega t + \pi/2)$

3. Consider a usual set up of Young's double slit experiment with slits of equal intensity as shown in the figure. Take 'O' as origin and the Y axis as indicated. If average intensity between $y_1 = -\frac{\lambda D}{4d}$ and $y_2 = +\frac{\lambda D}{4d}$ equals 'n' times the intensity of maxima, then 'n' equals



   (A) $\frac{1}{2}\left(1+\frac{2}{\pi}\right)$
   (B) $2\left(1+\frac{2}{\pi}\right)$
   (C) $\left(1+\frac{2}{\pi}\right)$
   (D) $\frac{1}{2}\left(1-\frac{2}{\pi}\right)$

Space for rough work

4. One of the combination from the fundamental physical constants is hc/G. The unit of this expression is
(A) kg^2 (B) m^3
(C) s^{-1} (D) none of the above

5. Let $[\varepsilon_0]$ denote the dimensional formula of the permittivity of vacuum. If M = mass, L = length, T = time and I = electric current. Then
(A) $[\varepsilon_0] = [M^{-1}L^{-3}T^2I]$ (B) $[\varepsilon_0] = [M^{-1}L^{-3}T^4I^2]$
(C) $[\varepsilon_0] = [M^{-1}L^2T^{-1}I^{-2}]$ (D) $[\varepsilon_0] = [M^{-1}L^2T^{-1}I]$

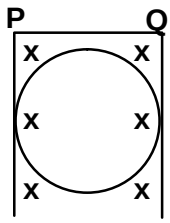
6. A coil of resistance R and inductance L is connected across an a.c. power supply of r.m.s. voltage V. The average power dissipated in the coil is,
(A) V^2/R (B) $\dfrac{V^2R}{(R^2+\omega^2L^2)}$
(C) $\dfrac{V^2}{\sqrt{(R^2+\omega^2L^2)}}$ (D) zero

7. The figure shows a plane wave front at a time 't' and at a time 't_1'
In the time interval (t_1 – t) the wave front must have passed through



(A) a prism (B) a prism and a convex lens
(C) a convex lens (D) a plane mirror and a concave lens

8. A vertical ring of radius r and resistance R falls vertically. It is in contact with two vertical rails which are joined at the top. The rails are without friction and resistance. There is a horizontal uniform magnetic field of magnitude B perpendicular to the plane of the ring and the rails. When the speed of the ring is v, the current in the section PQ is



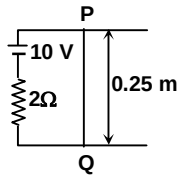
(A) zero (B) $\dfrac{2Brv}{R}$
(C) $\dfrac{4Brv}{R}$ (D) $\dfrac{8Brv}{R}$

Space for rough work

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 9 to 10

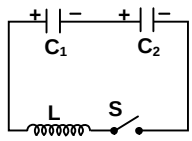


A metal wire PQ slides on parallel metallic rails having separation 0.25 m, each having negligible resistance. There is a 2 Ω resistor and 10 V battery as shown in figure.
There is a uniform magnetic field directed into the plane of the paper of magnitude 0.5T. A force of 0.5 N to the left is required to move wire PQ with constant speed to the right.
Then find the answer of following questions based on above paragraph.

9. With what speed is the wire PQ moving?
   (A) 8 m/s (B) 4 m/s
   (C) 16 m/s (D) 32 m/s

10. Taking this as a simple motor, what is its efficiency?
   (A) 0.3 (B) 0.2
   (C) 0.1 (D) 0.4

Paragraph for Question Nos. 11 to 12



In the figure shown C_1 = 1F, C_2 = 2F and L = 5H. Initially C_1 is charged to 50 V and C_2 to 10 V. Switch S is closed at time t = 0. Suppose at some instant charge on C_1 is 20 C with the same polarities as shown in the figure.

Then find the answer of following questions based on above paragraph.

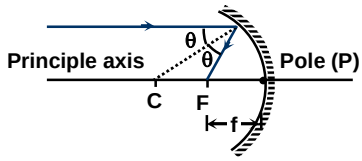
11. Energy stored in capacitor C_2 at this instant will be:
   (A) 10 J (B) 15 J
   (C) 25 J (D) 40 J

12. Current in the circuit at this instant will be
   (A) $10\sqrt{2}$ A (B) $15\sqrt{2}$ A
   (C) 10 A (D) 20 A

Space for rough work

Paragraph for Question Nos. 13 to 14

Spherical aberration in spherical mirrors is a defect which is due to dependence of focal length 'f' on angle of incidence 'θ' as shown in figure is given by



$$f = R - \frac{R}{2}\sec\theta$$

Where R is radius of curvature of mirror and θ is the angle of incidence. The rays which are closed to principle axis are called paraxial rays and the rays far away from principal axis called marginal rays. As a result of above dependence different rays are brought to focus at different points and the image of a point object is not a point.

Then find the answer of following questions based on above paragraph.

13. If f_P and f_m represent the focal length of paraxial and marginal rays respectively, then correct relationship is:

(A) $f_p = f_m$ (B) $f_p > f_m$

(C) $f_p < f_m$ (D) none

14. If angle of incidence is 60°, then focal length of this ray is:

(A) R (B) R/2

(C) 2R (D) 0

Multiple Correct Choice Type

This section contains **6 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

15. Suppose the potential energy between electron and proton at a distance r is given by $-\frac{Ke^2}{2r^3}$.

Using Bohrs theory choose the correct statements.

(A) Energy in the nth orbit is proportional to n^3

(B) Energy in the nth orbit is proportional to n^6

(C) Energy is proportional to m^2 (m : mass of electron)

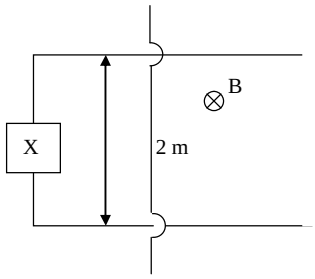
(D) Energy is proportional to m^{-3} (m : mass of electron)

Space for rough work

16. Let $[\varepsilon_o]$ denote the dimensional formula of the permittivity of the vacuum, and $[\mu_o]$ that of the permeability of the vacuum. If M = mass, L = length, T = time and I = electric current,
(A) $[\mu_o] = M^{-1}L^{-3}T^2I$ (B) $[\varepsilon_o] = M^{-1}L^{-3}T^4I^2$
(C) $[\mu_o] = MLT^{-2}I^{-2}$ (D) $[\mu_o] = ML^2T^{-1}I$

17. Which of the following (referred to a spherical mirror) do (does) not depend on whether the rays are paraxial or not?
(A) Pole (B) Focus
(C) Radius of curvature (D) Principal axis

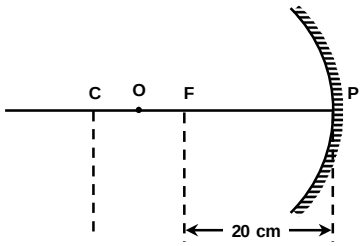
18. A resistance-less rod of length 'ℓ' and mass 'm' can slide without friction on long rails of zero resistance which are connected to circuit element named 'X' as shown. This arrangement is in the horizontal plane in a uniform vertical magnetic field 'B' as shown below. Then match what happens to the rod



(A) X is resistor and rod is thrown with some velocity towards right and it comes to rest
(B) X is a resistor and the rod is pulled to right with constant force and it attains terminal velocity
(C) X is resistor and rod is thrown with some velocity towards right and it never comes in rest
(D) X is a resistor and the rod is pulled to right with constant force and it never attains terminal velocity

19. A long wire carrying a steady current I lies in the plane of a circular conducting loop placed at a certain distance from the wire. There will be an induced current in the loop if it is
(A) moved parallel to the wire
(B) rotated about an axis perpendicular to the plane of the loop, passing through the centre of the loop
(C) rotated about an axis parallel to the wire passing through the centre of the loop
(D) moved away from the wire

20. A point object 'O' is kept is front of a concave mirror of focal length 20 cm at a distance of 30 cm from the pole. The object is moving on the principal axis away from the pole with a velocity of 1 cm/second. The instantaneous velocity of the image is



(A) 2 cm/sec towards the pole
(B) 4 cm such that the image distance tends to increase
(C) 4 cm away from the pole parallel to the principal axis
(D) 4 cm/sec towards the pole and parallel to the principal axis

Space for rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Aqueous solutions of $Na_2S_2O_3$ on reaction with Cl_2 gives
   (A) $Na_2S_4O_6$ (B) $NaHSO_4$
   (C) NaCl (D) NaOH

2. Both $[Ni(CO)_4]$ and $[Ni(CN)_4]^{2-}$ are diamagnetic. The hybridization of nickel in these complexes, respectively, are
   (A) sp^3, sp^3 (B) sp^3, dsp^2
   (C) dsp^2, sp^3 (D) dsp^2, dsp^2

3. At a certain temperature the dissociation constants of formic acid and acetic acid are 1.8×10^{-4} and 1.8×10^{-5} respectively. The concentration of acetic acid solution in which the hydrogen ion has the same concentration as in 0.001 M formic acid solution is equal to
   (A) 0.001 M (B) 0.01 M
   (C) 0.1 M (D) none of these

4. The increasing order of thermal stability for following salts will be
   (A) $FeC_2O_4 < CaC_2O_4 < CaCO_3 < K_2C_2O_4$ (B) $K_2C_2O_4 < CaCO_3 < FeC_2O_4 < CaC_2O_4$
   (C) $FeC_2O_4 < K_2C_2O_4 < CaC_2O_4 < CaCO_3$ (D) $CaCO_3 < K_2C_2O_4 < CaC_2O_4 < FeC_2O_4$

5. Which one of the following pairs of ions have the same electronic configuration?
   (A) Cr^{3+}, Fe^{3+} (B) Fe^{3+}, Mn^{2+}
   (C) Mn^{2+}, Fe^{2+} (D) Sc^{3+}, Cr^{3+}

6. A mono atomic gas is suddenly compressed to 1/8 of its volume adiabatically. The pressure of gas will change to
   (A) 24/5 times (B) 8 times
   (C) 40/3 times (F) 32 times

7. How do we differentiate between Fe^{3+} and Cr^{3+} in group III?
   (A) By adding excess of NH_4OH solution (B) By increasing NH_4^+ ion concentration
   (C) By decreasing OH^- ion concentration (D) Both (B) and (C)

Space for rough work

8. The compound which does not respond to ring test is
(A) $Pb(NO_3)_2$ (B) $LiNO_3$
(C) $NaNO_3$ (D) KNO_3

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for question Nos. 9 to 10

$(A) + NaCl \longrightarrow B(\text{white ppt.})$

$(B) + KI \longrightarrow C(\text{green ppt.})$

$(C) + KI(\text{excess}) \longrightarrow (D) + (E)(\text{colourless solution})$

$(E) + NH_3 + KOH \longrightarrow F$

9. Compound (A) and (B) are respectively
(A) $AgNO_3$ and $AgCl$ (B) $Pb(NO_3)_2$ ad $PbCl_2$
(C) $Hg_2(NO_3)_2$ and Hg_2Cl_2 (D) $Cu_2(NO_3)_2$ and Cu_2Cl_2

10. When compound (B) and reacts with NH_3 solution, the colour of the compound formed is
(A) black (B) red
(C) yellow (D) white

Paragraph for question Nos. 11 to 12

The binary compounds of oxygen with other elements are called oxides. They are classified either depending upon their acid – base characteristics or on the basis of oxygen content.

11. Which of the following pairs contains mixed oxides?
(A) Pb_3O_4, Fe_3O_4 (B) MnO_2, BaO_2
(C) KO_2, Na_2O_2 (D) Mn_3O_4, N_2O_5

12. Which of the following pairs contain amphoteric oxides?
(A) BeO, BaO (B) BeO, Al_2O_3
(C) Al_2O_3, P_2O_5 (D) FeO, CuO

Space for rough work

Paragraph for Question Nos. 13 to 14

A metal is regarded as an assembly of metal ions and free electrons. When the metal is in contact with water, some metal ions enter into the liquid due to a tendency in the metal, called by Nernst as "Electrolytic solution tension". An alloy of 1.45 gm Ag and Si was dissolved in desired amount of HNO_3 and volume made upto 500 ml. An Ag electrode was dipped in solution and E_{cell} of the cell is $Pt\ \ H_2\ |H^+\ ||\ Ag^+/Ag$ was 0.503 V at 298 K. $\left[E^o_{Ag^+/Ag} = 0.80\ V\right]$

(1 atm) (1 M)

Read the above paragraph carefully and answer the questions given below:

13. Observed concentration of Ag^+ in the cell
(A) 2.25×10^{-9} M (B) 9.25×10^{-6} M
(C) 9.25×10^{-7} M (D) 9.25×10^{-5} M

14. The emf of the cell involving the reaction, $2Ag^+(aq) + H_2(g) \longrightarrow 2Ag(s) + 2H^+(aq)$, is 0.80 V. The standard oxidation potential of Ag electrode is
(A) 0.80 V (B) –0.80 V
(C) 0.40 V (D) 0.20 V

Multiple Correct Choice Type

This section contains **6 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

15. Which of the following exhibit Frenkel defects?
(A) AgI (B) AgCl
(C) KBr (D) ZnS

16. A d-block element forms octahedral complex but its magnetic moment remains same either in strong field or in a weak field ligand. Which of the following is/are correct?
(A) Element always form colourless compound
(B) Number of electrons in t_{2g} orbitals are higher than in e_g orbitals
(C) It can have either d^3 configuration
(D) It can have either d^7 configuration

17. The compound (s) which have peroxo linkage is (are)
(A) $H_2S_2O_3$ (B) H_2SO_5
(C) $H_2S_2O_7$ (D) $H_2S_2O_8$

Space for rough work

18. Identify the incorrect statement(s) among the following?
(A) CN^- is a pseudo halide
(B) Cl_2O(gas) on dissolving in KOH solution produces $KClO_3$
(C) Bleaching action of chlorine occurs in presence of moisture
(D) $H_3P_3O_9$ contains three P–P bonds

19. For a cell reaction represented by the equation
$Cu(s) + 2Ag^+(aq) \longrightarrow Cu^{2+}(aq) + 2Ag(s)$
The value of ΔG^o could be obtained by
(A) setting up $Cu(s)/Cu^{2+}(aq)$ and $Ag(s)/Ag^+(aq)$ half-cells (using molar solution and measuring the emf of the resulting cell)
(B) adding powdered copper to a solution of Ag^+ ions and determining the enthalpy change, repeating this experiment at different temperatures
(C) setting up $Cu(s)/Cu^{2+}(aq)$ and $Ag(s)/Ag^+(aq)$ half-cells and determining the molarities of the ions which gives an emf of zero for the cell
(D) setting up $Cu(s)/Cu^{2+}(aq)$ and $Ag(s)/Ag^+(aq)$ half-cells, using the resultant emf to drive a motor and measuring the work which it can do

20. In the depression of freezing point experiment, it is found that the
(A) vapour pressure of the solution is less than that of pure solvent
(B) vapour pressure of the solution is more than that of pure solvent
(C) only solute molecules solidify at the freezing point
(D) only solvent molecules solidify at the freezing point

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **8 multiple choice questions** numbered 1 to 8. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. Find the area enclosed between $x^2 + y^2 \le 9\pi^2$ and $\sin(x - y) \ge 0$

(A) $\frac{9\pi^3}{4}$ (B) $\frac{9\pi^3}{2}$

(C) $3\pi^3$ (D) π^3

2. Consider an invertible function f(x) with $f'(x) > 0$ and $f''(x) < 0 \ \forall \ x \in R$. Also $f^{-1}(x)$ is the inverse function of f(x). Let f(x) and $f^{-1}(x)$ intersect each other at x = 0 and $x = \alpha(\alpha > 0)$ only and if $\int_0^{\alpha} f^{-1}(x)dx = A$; $\int_0^{\alpha} f(x)dx = B$ and $\frac{\alpha^2}{2} = C$ then

(A) A > B > C (B) A > C > B

(C) A < B < C (D) A < C < B

3. The area bounded by the curve $y^2 \le 4x$ and the circle $x^2 + y^2 - 2x - 3 \le 0$ is equal to

(A) $2\pi + \frac{8}{3}$ (B) $4\pi + \frac{8}{3}$

(C) $\pi + \frac{8}{3}$ (D) $\pi - \frac{8}{3}$

4. If P^{th}, q^{th}, r^{th} term of a GP are the positive numbers a, b, c then angle between the vectors $\log a^3\hat{i} + \log b^3\hat{j} + \log c^3\hat{k}$ and $(q-r)\hat{i} + (r-p)\hat{j} + (p-q)\hat{k}$ is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$ (D) $\sin^{-1}\left(\frac{1}{\sqrt{a^2+b^2+c^2}}\right)$

5. If A is a square matrix of order n such that $|\text{adj}(\text{adj}A)| = |A|^9$, then the value of n can be

(A) 1 (B) 2

(C) 3 (D) 4

Space for rough work

6. Given that $A=\begin{bmatrix}1 & 0 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix}$ and $A^n = A^{n-2} + A^3 - A \ \forall n \geq 4$ and $A^3 - A^2 - A + I = 0$.

$A^{20}=\begin{bmatrix}1 & 0 & 0\\ k+1 & 1 & 0\\ k+1 & 0 & 1\end{bmatrix}$. Then k is equal to

(A) 3 (B) 6
(C) 9 (D) none of these

7. Let A be the event that P (A) = 0, then
(A) A is impossible event (always) (B) A could be possible
(C) A can be said (D) none of these

8. Forces $2\hat{i}+7\hat{j}, 2\hat{i}-5\hat{j}+6\hat{k}, -\hat{i}+2\hat{j}-\hat{k}$ act at a point P whose position vector is $4\hat{i}-3\hat{j}-2\hat{k}$. Then the vector moment of the resultant of three forces acting at P about the point Q, where position vector is $6\hat{i}+\hat{j}-3\hat{k}$.

(A) $-24\hat{i}-13\hat{j}-4\hat{k}$ (B) $24\hat{i}-13\hat{j}-2\hat{k}$
(C) $-24\hat{i}+13\hat{j}+4\hat{k}$ (D) none of these

Comprehension Type

This section contains **3 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONLY ONE** is correct

Paragraph for Question Nos. 9 to 10

Read the following write up carefully and answer the following questions:

Let $M_1=\begin{bmatrix}3 & -4\\ 1 & -1\end{bmatrix}$ and $M_2=\begin{bmatrix}x & 4\\ 0 & 1\end{bmatrix}$ ($x \neq 0, 1, x \in R$) are two matrices and let $P=\begin{bmatrix}\frac{\sqrt{3}}{2} & \frac{1}{2}\\ -\frac{1}{2} & \frac{\sqrt{3}}{2}\end{bmatrix}$ any matrix.

Also matrix $A = PM_1P^T$, $B = P^TA^{2012}P$, $C = PM_2P^T$ and $D = P^TC^{2012}P$.
Now answer the following questions

9. Trace (B + D) = 0 for
(A) x = 2 (B) x = –1
(C) more than 100 values of x (D) no value of x

Space for rough work

10. Let matrix $E = (M_1B^{-1}M_2D^{-1})^{-1}$, then sum of all elements of second row of E is equal to
(A) 0 (B) 2012
(C) -2010 (D) none of these

Paragraph for Question Nos. 11 to 12

Read the following write up carefully and answer the following questions:
If $P(A_1 \cap A_2) = P(A_1).P(A_2)$ then A_1, A_2 are called independent events for each pair is A_1, A_2, A_3 if $(A_i \cap A_j) = P(A_i).P(A_j)$ then A_1, A_2, A_3 are called mutually pair-wise independent events

11. An contains four tickets, each is marked as aab, aba, baa, bbb. And one ticket is drawn. Let A_i = event that i^{th} digit of the mark on ticket is a. Then
(A) A_1, A_2, A_3 are pair-wise independent
(B) A_1, A_2, A_3 are mutually independent
(C) A_1, A_2, A_3 are need not be pair-wise independent
(D) none of these

12. A and B are two independent events. The probability that both occur simultaneously is $\frac{1}{6}$ and the probability the neither occurs is $\frac{1}{3}$, then
(A) $P(A) = \frac{1}{2}$, $P(B) = \frac{1}{3}$ (B) $P(A) = \frac{2}{3}$, $P(B) = \frac{1}{3}$
(C) $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{3}$ (D) none of these

Paragraph for Question Nos. 13 to 14

Read the following write up carefully and answer the following questions:
If P(x,yz) is varying such that $ax + by + cz + d = 0$ where $a^2 + b^2 + c^2 \neq 0$ then P traces a plane with normal $a\hat{i} + b\hat{j} + c\hat{k}$ and $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$ gives plane equation passing through $A(x_1, y_1, z_1)$ with normal vector $a\hat{i} + b\hat{j} + c\hat{k}$ $(a^2 + b^2 + c^2 \neq 0)$

13. A variable plane with a constant distance 'd' from origin cuts x, y, z axes at P, Q, R. Then locus of centroid of triangle PQR is
(A) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{d^2}$ (B) $\frac{1}{x^3} + \frac{1}{y^3} + \frac{1}{z^3} = \frac{9}{d^3}$
(C) $\frac{1}{\sqrt{|x|}} + \frac{1}{\sqrt{|y|}} + \frac{1}{\sqrt{|z|}} = \frac{1}{\sqrt{|d|}}$ (D) $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{9}{d^2}$

Space for rough work

14. A variable plane at a constant d from origin, intersects the axes at A, B, C. Planes are drawn to the coordinate planes and passing through A, B, C. Then locus of point of intersection of all planes is

(A) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{1}{p^2}$ (B) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{2}{p^2}$

(C) $\frac{1}{x^2}+\frac{1}{y^2}+\frac{1}{z^2}=\frac{3}{p^2}$ (D) none of these

Multiple Correct Choice Type

This section contains **6 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct

15. Given three distinct planes $p_1: ax+by+cz=1$, $p_2: bx+cy+az=1$, then $p_3: cx+ay+bz=1$

(A) if the planes intersect at the single point (3,3,3) then a + b + c = $\frac{1}{3}$

(B) if a + b + c= 0, then the planes will not intersect at a common point

(C) if $a+b+c\neq 0$ then the lines $\frac{x-1}{a}=\frac{y}{b}=\frac{z+1}{c}$ and $\frac{x+1}{c}=\frac{y}{b}=\frac{z-1}{a}$ will never intersect

(D) the lines $\frac{x-1}{a}=\frac{y-1}{b}=\frac{z-1}{c}$ and $\frac{x+1}{b}=\frac{y+1}{c}=\frac{z+1}{a}$ will never intersect

16. Which of the following is/are correct?
(A) P is orthogonal iff P^T is orthogonal
(B) if P is orthogonal, then P^{-1} is orthogonal
(C) if P and Q are orthogonal, then PQ is orthogonal
(D) none of these

17. Let A denotes the matrix $\begin{bmatrix}0 & i\\ i & 0\end{bmatrix}$, where $i^2=-1$

(A) $I + A + A^2 + ... + A^{2012} = I$ (B) $A^{4n+1}= A, n \in N$

(C) $A^{4n+2}= I, n \in N$ (D) $I + A + A^2 + ... + A^{2012} = A$

18. If A, B, C are the angles of a triangle and $\Delta=\begin{vmatrix}\sin^2 A & \sin A\cos A & \cos^2 A\\ \sin^2 B & \sin B\cos B & \cos^2 B\\ \sin^2 C & \sin C\cos C & \cos^2 C\end{vmatrix}=0$, then

(A) triangle must be equilateral (B) triangle must be isosceles

(C) triangle can be equilateral (D) triangle must be right angled

Space for rough work

19. Let the vectors $\alpha\hat{i}+3\hat{j}-\hat{k},\ \hat{i}+(\alpha-1)\hat{j}+2\hat{k},\ 3\hat{i}+5\hat{j}+2\hat{k}$. Then

(A) the number of planes determined above 3 vectors = 2

(B) angle between planes $=\cos^{-1}\left(\frac{17}{3\sqrt{23}}\right)$

(C) angle between planes $=\tan^{-1}\left(3/\sqrt{5}\right)$

(D) none of these

20. Let two curves y = f(x) with f(0) = 1 and $g(x)=\int_{-\infty}^{x} f(t)dt$ with g(0) = 4 are such that, their tangents at the points with equal abscissa intersects on x-axis, then

(A) $\int_{-1}^{1}\frac{x^2+4}{4+g(x)}dx=\frac{13}{3}$

(B) area bounded by y = f(x), y = g(x), x = 1 and y axis is $12(e^{1/4}-1)$ sq. unit

(C) f(x) – g(x) is decreasing in [–1, 1]

(D) period of $f\left(\left\{\frac{x}{2}\right\}\right)$ is 8 ({.} denotes fractional part of x)

Space for rough work